Hazard Identification and Inspection Protocol Sheet

Performing Arts (CU 761030)

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Table of Contents

Performing Arts (CU 761030)....4
Industry or Classification Unit Information....4
Industry or Classification Unit Hazards....5
Prevention Officer Personal Safety While On-Site....14
Risk Assessment Process....14
Personal Protective Equipment (for Workers)....15
Regulations to Consider....15
Additional Sources of Information....17

Performing Arts (CU 761030)

Industry or Classification Unit Information

Name of Industry or Classification Unit	Performing Arts – CU #761030
Description of typical work conducted in this Industry or Classification Unit	This classification unit covers both firms that are responsible for all aspects of live productions and independent performers. Live productions may include bands, orchestras, touring companies or circuses without animals. Some examples of services that firms in this classification unit provide are as follows: • Disc jockey service • Keynote speaker • Perform live art • Karaoke service • Musical service • Produce live performances Some examples of equipment that firms in this classification unit use are as follows: • Orchestral instrument • Stage light & sound equipment Some examples of key workers in this classification unit are as follows: • Actor • Conductor • Director • Entertainer • Production manager • Set builder • Special effects person • Choreographer • Dancer • Disc jockey • Musician • Pyrotechnician • Set designer

	• Stage crew

Industry or Classification Unit Hazards

Does this industry or classification unit contain hazards relating to **confined spaces/confined space entry**?
Orchestra pit lift area, elevator pits, house cover lighting, under stages and flooring. **NOTE**: These may also meet the definition of an enclosed space; however may meet definition if dry ice is also being used and not vented. The confined spaces are typically what you would find in facilities or surrounding area for where the live performances are taking place (e.g., boilers, manholes, air conditioning ducts, etc.)
Does this industry or classification unit contain hazards relating to **working from heights**?
Work at heights is typically done when striking, theatre or arena rigging (high steel), or during performance (for example performer flying and aerial stunts). Fall hazards are present when working on ladders, on the catwalks, outside of the catwalks, and on unprotected elevated work platforms such as the open edge of a stage. Ladders, lifts, scaffolds, work platforms, balconies, catwalks, stage, stilts, scenery, flying rigs, tension grids, aerial work platforms, or other elevated work surfaces are used. Working above an audience, an orchestra pit, or on platforms or risers. All flying system operators and performers including understudies must be fully trained by a qualified person and rehearsed with the flying equipment. The equipment used is often attached to the stationary structural members of a building or theatre, to a movable batten of a mechanical counterweight or motorized rigging system, or to a temporary truss or supporting member. The performing arts presents unique difficulties when it comes to working at heights. For example, guardrails are often not practicable on scenic units that will be visible to an audience; similarly, full body harnesses and safety lines on performers are often not practical during performance. Use of tension grids to provide access to theatrical lighting, audio cables, loudspeakers, and stage rigging. Any edge of the grid that is not attached directly to a wall must be equipped with guardrails.

Does this industry or classification unit contain hazards relating to **machine safeguarding**?
Required for the use of any power tools for set or prop construction.
Does this industry or classification unit contain hazards relating to **machine and equipment lockout**?
Limited to electrical energy. However, pneumatic energy should also be considered, as the use of air may require electrical but there may be a hazard of stored pneumatic energy present. Electrical equipment of various types (from machines used to create prop/set construction to lighting, to sound equipment). Use of electrical in wet conditions; required use of ground fault circuit interrupters GFCIs and residual current device (RCD) protection.
Does this industry or classification unit contain hazards requiring **respiratory programs**?
Chemicals associated with construction or special effects (e.g., smoke and fog)
Does this industry or classification unit require **Exposure Control Plans (ECPs)**?
Glycol or other chemicals maybe used for special effects. Heat and cold stress ECPs may be required. Depending if performances are outside or could be working inside with high activity level and wearing costumes that do not breathe. Bloodborne pathogen ECP for first aid attendants. If welding activities are being conducted, an ECP regarding welding fumes may be required.
Does this industry or classification unit contain hazards relating to **mobile equipment and/or the requirement for roll-over protection (ROPs)**?
Types of lifts (fork lift, scissor, boom, snorkel lifts, etc.) Qualified personnel co-ordinate and train performers/workers in use of lifts. Insert-camera cars may be used for any event that is being filmed.

Does this industry or classification unit contain hazards of **musculoskeletal injuries (MSI)**? If “Yes” please explain/describe them, in detail, specific to the industry or classification unit
Back strains from manual material handling, particularly when setting up and striking. Sprains and strains by performers especially in dancers. Prolonged sitting and awkward postures for example sitting in orchestra pits. Stage pieces are often awkward, heavy, or oddly shaped, which makes them difficult to lift properly.
Does this industry or classification unit contain hazards of **working alone**?
Not generally applicable. Performers could be rehearsing alone.
Does this industry or classification unit contain hazards relating to **contractor safety**?
The use of independent contractors is a typical practice. Contractors may be used to construct sets, props, rigging, special effects, etc. Cast and crew can also be independent contractors, including directors, performers, stunt men/women, etc.

Does this industry or classification unit contain other specific hazards not previously listed?	
Type of hazard	**Description of hazard**
Scaffolds	Scaffolds can be used when set construction crews build sets, or can be used sometimes as a set piece(s).
Rigging	Generally refers to anything that is used for attaching, supporting or flying scenery, drapery, lighting, audio, visual and sound equipment. This includes fibre ropes, wire ropes, chains, slings, attachments, connecting fittings and associated components. Rigging is the hardware to lift, lower, and hold performance equipment on or above a stage. A variety of rigging hardware may be used for various tasks, and understanding the load capacity for each piece of equipment is critical.

Does this industry or classification unit contain other specific hazards not previously listed?	
Type of hazard	**Description of hazard**
	Rigging presents hazards for: • The operator (musculoskeletal and other bodily injuries) • The performers or crew onstage below (being crushed by falling scenery) • The audience (being crushed by falling scenery) • Other set pieces, furniture, stage floors and the rigging equipment itself. Hoists and winches are used to move all types of materials and equipment such as scenery, platforms, flying individuals, operating fire safety curtains, moveable lighting instruments, and hoisting trusses. Ropes and cordage are used to operate the rigging mechanisms and to hoist materials and/or equipment to overhead platforms.
Ladders	Extension trestle ladders mounted on a castered base are commonly used, particularly on theatre stages where access and weight may be an issue. Workers on these ladders use what is called a “safe stance” – straddle the top rung and place one foot on either side on the second or third rungs. Keep your centre of gravity close to the ladder’s centre line. A common industry practice that should be stopped. Fixed ladders permanently attached to a structure, building or equipment. Used to access elevated areas, such as fly lofts, the tension grid, and bridges/catwalks. Step ladders are used frequently in set-up, production, and striking.
Slips, Trips, and Falls (i.e., workplace conditions)	Lighting and other electronics are commonly used. A lot of planning goes into placement and the running of cables. Improperly run cables can lead to housekeeping issues that pose trip and fire hazards. Cable covers are a frequent cause of tripping hazards. Temporary electrical wiring must be clearly marked and secured on floors to prevent tripping.

Does this industry or classification unit contain other specific hazards not previously listed?	
Type of hazard	**Description of hazard**
	Scenery such as elevated performance stages and scenic units may be permitted for workers to stand; can cause trip hazards if low lighting or no hand rails. Pits, trap doors, and changes in elevation (stairs, ladders, etc.) pose trip and fall hazards. Due to aesthetics, controls may not be practical.
Electrical	Can be found throughout operations and include, but are not limited to, exposed wiring; improperly spliced wires; improperly installed temporary power distribution; improperly grounded wiring; use of damaged electrical cords; i.e. frayed, repaired, missing ground prongs, and the use of inappropriate extension cords. Lighting instruments can be plugged into raceways that are attached to line sets that are commonly referred to as electrics. The raceways contain many wires that when energized are dangerous. Electronics must not be close to flammable materials, such as scenery, draperies, etc. Proper grounding of audio equipment is necessary. For outdoor venues or performances these but be protected from inclement weather and the use of appropriate equipment for working in wet conditions.
Set interaction	Interaction between members of cast, crew, musicians and audience while on stage. Stage lifts, holes, openings, pits, revolves, traps and elevated areas. Crew or performers being in the wrong place on stage at the wrong time (missing marks).
Inappropriate performance surfaces	Inadequately supported floors. Inappropriate surfaces for dancers, such as floor that are not sprung.
Moving objects	Moving scenery flown incorrectly or unsafely. Scene changes. Placement of props near unprotected edges.

Does this industry or classification unit contain other specific hazards not previously listed?	
Type of hazard	**Description of hazard**
Costume design and performance	Costumes can present hazards both to the cast members wearing them and to the wardrobe attendants responsible for cleaning and maintaining them. In addition, the making of special costumes such as armor, masks, etc. can be hazardous due to the type of materials (for example, plastic resins, foams, etc.) used. Special costumes involving metal, wood frameworks, etc. could be heavy and hot for the person wearing them. Costumes should be non-irritant, comfortable, ventilated, simple to maneuver in, and easy to communicate through. In addition to the wearer, other hazards to consider include manual materials handling and ergonomics for the costume designer as well as any wardrobe attendants. Costumes and theatrical makeup can increase the potential for a heat stress disorder.
Theatrical makeup	Makeup includes a wide range of products with a variety of ingredients, including solvents, dyes, pigments, preservatives, oils, waxes, etc. Some of these ingredients can cause health effects in people, such as skin or eye irritation or allergic reactions. Some ingredients are a known or probable human carcinogen. Inhalation of solvents from hair sprays and other spray products can also be hazardous. A potentially hazardous act in the industry is the sharing of makeup. A single makeup artist may work on multiple people daily which could cause the transmission of bacterial infections and communicable diseases. Special effects makeup using adhesives, sealers, removers, paints and various facial and body appliances, including dentures, and contact lenses, etc. are used to change the appearance of performers. Products such as liquid latex, gelatin, foam latex, silicone rubber molding compounds, polyurethane rubber and foams, etc. are used to make different types of prosthetic appliances.

Does this industry or classification unit contain other specific hazards not previously listed?	
Type of hazard	**Description of hazard**
	Other potentially hazardous acts and conditions are the removal of special effects makeup. Products such as solvents, acetone, mineral spirits, isopropyl myristate, etc. may be used on skin.
Lighting	Performing arts uses a lot of electricity and the risk of electrocution is high. Conducting operations such as hanging lights, sometimes in the dark, with high-voltage electricity has the potential to cause a variety of incidents such as falls, fires, electrocution, and injuries from falling objects. In addition, lighting equipment is heavy and can cause injury if proper lifting techniques are not used. For cast members moving through different light levels from very bright stage lighting to dim backstage lighting can be hazardous. Often there may be no backstage running lights. Lighting rigs are very heavy and can cause injury if they fall; lights must be properly hung and secured. Followspot lighting is common practice. A light that is physically moved by an operator to follow a performer. Must be checked for electrical attachments, worn cables, grounding, gel frame, stability of the unit, lamp replacement, etc.
Fatigue	Long rehearsals. Insufficient rest and food breaks. High levels of physical exertion or unusual or awkward physical postures or activities.
Manual handling	Particularly an issue when striking. Mechanical lifting devices may not be an option depending on layout and surface of set. Loading and unloading material from trucks, trailers, and storage shelves presents unique risks. Serious injuries are common during loading and unloading operations.

Does this industry or classification unit contain other specific hazards not previously listed?	
Type of hazard	**Description of hazard**
Violence	Increased risk in large public area or where alcohol is served. Illegal drugs being brought and consumed in working environment.
Radiation – non-ionizing	Use of lasers in light shows.
Explosives and pyrotechnics	Prematurely triggering the pyrotechnic effect, use of larger quantities or more dangerous materials than needed, causing a fire, lack of adequate fire extinguishing capabilities, and inadequately trained and experienced pyrotechnic operators. Fire permits are required in most jurisdictions.
Special props	Some events or performances call for special props, such as weapons or live animals. These props pose unique risks that require special handling procedures to ensure the safety of performers, crew, and audience. Artificial snow may be made from chemical mixes, shredded paper, or plastic. Chemically produced snow may pose hazards such as eye and skin irritation or internal ingestion risks. May also be a fire hazard.
Strike	Deconstruction and removal of stage, sets, props, costumes, lights, dressing stations, catwalks, and sound equipment. The deconstruction of shows in a production house often involves recycling and reusing as much of the material as possible. The strike is busy and loud and includes manual material handling.
Smoke and fog (special effects)	Smoke is an atmospheric effect composed of solid particulate produced by combustion, where fog or haze is composed of liquid droplets. Smoke is produced using pyrotechnic materials, such as white or color smoke cartridges, or other flammable substances. Fog or haze can be produced using a variety of commercially available machines using glycol, glycol/water mixtures, water, mineral oil, or dry ice. Air monitoring may be required to ensure workers are not exposed to levels above acceptable limits.

Does this industry or classification unit contain other specific hazards not previously listed?	
Type of hazard	**Description of hazard**
	The use of atmospheric smoke, fog and/or haze may change a non-hazardous confined space into a hazardous confined space.
Ergonomics	Performing arts work involves a lot of lifting and moving of items. Improperly lifting items increases risk of injury.
Front of house	Fire and life safety. Emergency evacuation plans must include front of house personnel. Patron safety; begins prior to arrival and continues until departure from facility. Personnel are responsible for making safety announcements, crown control, safety rule enforcement, emergency response, and incident investigations.

Prevention Officer Personal Safety While On-Site

Does this industry or classification unit require specific **Personal Protective Equipment (PPE)** the Prevention officer must don while they are on-site?
PPE necessary will depend on the task and the work occurring around the officer. Hard hats – when overhead hazards are present Reflective vests – required to increase visibility or when near mobile equipment Hearing protection – when high noise levels are generated during rehearsals and productions Footwear – where there is a risk of falling objects, rolling objects, or objects that may pierce the sole of the shoe. Respiratory protection – more typically used when in prop shop or the creating of scenery, props If/when inspecting outdoors, appropriate sunscreen should also be used.
Does this industry or classification unit require the Prevention officer to use **air monitoring equipment** while they are on-site?
Not applicable
Does this industry or classification unit require the Prevention officer to protect themselves against **dogs** while they are on-site?
Not applicable

Risk Assessment Process

Does this industry or classification unit require a specific **Risk Assessment Process** the Prevention officer must/should follow while they are conducting an inspection?
Same as any other industry. Stunt work requires a specific stunt risk assessment process.

Personal Protective Equipment (for Workers)

Does this industry or classification unit require specific **Personal Protective Equipment** that the workers must don while they carry out their job functions?
PPE necessary will depend on the task and the work occurring around the worker. Hard hats – when overhead hazards are present Safety glasses and goggles – when using hand and power tools or there is a risk of flying debris or the use of lasers. Work gloves – when required to handle materials with abrasive edges Reflective vests – required to increase visibility or when working near mobile equipment Hearing protection – when high noise levels are generated during rehearsals and productions Footwear – where there is a risk of falling objects, rolling objects, or objects that may pierce the sole of the shoe. Respiratory protection – more typically used in prop shop or the creating of scenery, props, or applying theatrical makeup. May be used by crew where exposure to special effects (i.e., smoke and fog) exceed allowable limits. If/when conducting work activities outdoors, sunscreen should also be used.

Regulations to Consider

Hazard/Violation/Topic	Regulation Reference
Lack of written first aid and emergency response procedures.	OHSR Section 3.17(1), First aid *procedures*
Lack of guardrails on scaffold. The scaffold was used for set construction and lighting. The employer representative stated that the scaffold did not have the required guard rails for when working over fall hazards and at heights.	OHSR Section 13.15, *Manufactured components*
Fatigue/impairment.	OHSR Sections 4.19(1) and 4.19(2), *Physical or mental impairment*

Hazard/Violation/Topic	Regulation Reference
Unexpected energization or start-up of machinery or equipment, or unexpected release of an energy source.	OHSR Section 10.2, *General Requirement*
Guards on table saws not in place.	OHSR Section 12.59(1), *Removing guards*
No health and safety program where the workforce is less than that referred to in Section 3.1(1).	OHSR Section 3.2, *Small operations*
Exposed electrical wiring, uncovered electrical panel, and/or uncovered junction box.	OHSR Section 19.12(1), *Working close to energized equipment*
Tool, machine, piece of equipment is capable of safely performing the functions for which it is used and is selected, used and operated in accordance safe work practices and with the manufacturer's instructions.	OHSR Sections 4.3(1)(b)(i) and (ii), *Safe machinery and equipment*
Exposure to a substance that exceeds the ceiling limit, short-term exposure limit, or 8-hour TWA limit prescribed by ACGIH.	OHSR Section 5.48, *Exposure limits*
Exposure to a hazardous substance by any route of exposure, the employer must eliminate the exposure, or otherwise control it below harmful levels and below the applicable exposure limit established under section 5.48 by (a) substitution, (b) engineering control, or (c) administrative control, (d) personal protective equipment.	OHSR Section 5.55(1), *Type of controls*
The employer has failed to implement an exposure control plan to maintain worker exposures to designated substances as low as reasonably achievable below the exposure limit established under section 5.48.	OHSR Section 5.57(2), *Designated Substances*
Work is being done at a place from which a fall of 3 m (10ft) or more may occur.	OHSR Section 11.2(1)(a), *Obligation to use fall protection*
Ropes, chains, and other lines on scenery (i.e., flys)	OHSR Section 15.4(2), *Use of rigging* OHSR Section 15.6, *Design Factors*

Hazard/Violation/Topic	Regulation Reference
Stairways (treads and backstage stairs are in good condition, properly lit, alleyways are clear of litter and obstacles).	OHSR Section 4.65, *Illumination levels* OHSR Section 20.4(2), *Safe access*
SDS reference sheets available for reference.	OHSR Section 5.14, *Supplier SDS* OHSR Section 5.15, *Employer SDS* OHSR Section 5.16, *Availability of an SDS*
If black light is used, low in harmful ultra violet radiation.	OHSR Section 7.19(5), *Exposure Limits*
Storage of flown scenery.	OHSR Section 4.43, *Stacking materials* OHSR Section 4.43.1, *Storage racks*

Additional Sources of Information

Topic	Source
Health and Safety Association for the industry and employers in CU 761030	ActSafe (https://www.actsafe.ca/)
Health and Safety for Motion Picture and Television Production in B.C.	WorkSafeBC Publication